

Business Problem: A leading retail company seeks to understand customer shopping behavior to improve sales, satisfaction, and loyalty.

- **Key Challenge:** Changes in purchasing patterns across demographics, product categories, and sales channels (online vs. offline) have been observed.
- **Core Interest:** Uncovering factors (discounts, reviews, seasons, payment preferences) that drive consumer decisions and repeat purchases.
- **Overarching Business Question:** "How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"
- **Deliverables:**
 1. **Data Preparation & Modeling (Python):** Clean and transform the raw dataset.
 2. **Data Analysis (SQL):** Organize data, simulate transactions, and run queries for insights (segments, loyalty, purchase drivers).
 3. **Visualization & Insights (Power BI):** Build an interactive dashboard to highlight key patterns and support data-driven decisions.
 4. **Report and Presentation:** Summarize findings, provide business recommendations, and visually communicate insights to stakeholders.
 5. **GitHub Repository:** Include all Python scripts, SQL queries, and dashboard files.